

Synergistic Molecular Stacking in an Asymmetric Self-Assembled Bilayer for Universally Elevating the Performance of Perovskite Photovoltaics

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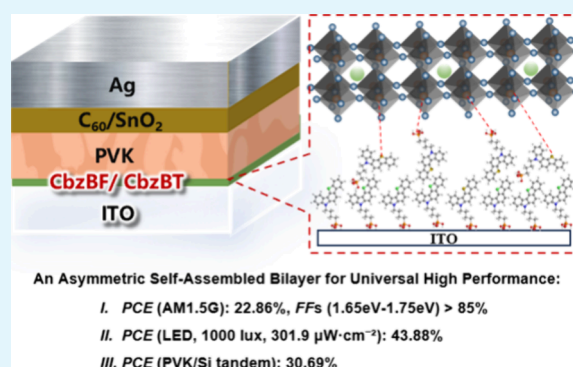
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ABSTRACT: Perovskite photovoltaics have demonstrated remarkable progress in power conversion efficiency, showing great potential for broad applications. However, interfacial issues such as energy-level mismatch and defect density remain key challenges that limit device performance. To address the rigid energy levels and inhomogeneous coverage of conventional symmetric self-assembled monolayers (SAMs), we present an asymmetric self-assembled bilayer strategy. In this design, a bottom CbzBF SAM is anchored to the ITO substrate to optimize energy-level alignment for hole extraction, while a top CbzBT SAM simultaneously passivates interfacial defects and modulates perovskite crystallization. As a result, single-junction perovskite photovoltaics (PPVs) with a bandgap of 1.68 eV attain power conversion efficiency (PCE) of 22.86% with a high fill factor (FF) of 86.54% under standard AM1.5G illumination, along with 43.88% PCE under 1000 lx indoor LED lighting. Critically, the CbzBF/CbzBT bilayer enabled fill factors over 85% in PPVs with bandgaps of 1.65, 1.72, and 1.75 eV, ranking among the highest reported values and underscoring its exceptional universality. Moreover, our two-terminal perovskite/silicon tandem solar cells reached a PCE of 30.69%. This study presents a practical and effective interfacial design strategy toward high-performance perovskite photovoltaics.

KEYWORDS: asymmetric bilayer, sequential deposition, wide-bandgap perovskites, high fill factor, tandem solar cells



INTRODUCTION

Due to their high absorption coefficients, long carrier diffusion lengths, and tunable bandgaps, perovskite photovoltaics (PPVs) have emerged as a revolutionary light-harvesting technology.^{1,2} In particular, inverted (p–i–n) structured wide-bandgap (WBG, $E_g \geq 1.65$ eV) PPVs show great promise for both tandem architectures and indoor photovoltaics.^{3–5} Under indoor light, WBG perovskites align well with the emission spectra of light-emitting diode (LED), enabling high open-circuit voltages (V_{OC}) and remarkable power conversion efficiencies (PCE), making them ideal for low-power electronic devices and Internet of Things (IoT) devices.^{6–9} In tandem configurations, bandgap-tunable perovskites streamline the optimization of top- and bottom-cell bandgap alignment, enabling efficient spectral splitting and broader solar spectrum utilization.^{10–13} Notably, two-terminal perovskite/silicon tandem cells have surpassed the Shockley–Queisser limit for single-junction devices, achieving a certified efficiency of 34.85%.¹⁴

The hole transport layer (HTL), especially self-assembled monolayers (SAMs), is critical for efficient hole extraction and perovskite crystallization.^{15,16} Carbazole-based SAMs, such as

2PACz, MeO-2PACz, Me-4PACz, and MeO-4PACz, are widely adopted in indoor and tandem perovskite photovoltaics. However, their symmetric molecular structures often result in limited dipole moments and suboptimal energy-level alignment.^{17,18} Recent efforts have turned to asymmetric SAMs to tailor energy levels and passivate deep-level defects in WBG perovskites.^{19,20} For instance, Kim et al.²¹ replaced a methoxy group with a fluorine atom in MeO-4PACz, enhancing the dipole moment, work function, and HOMO level depth, which led to certified PCEs of 20.14% for single-junction PPVs and 30.05% for perovskite/silicon tandems. Jen et al.¹⁹ introduced oxygen and sulfur atoms into a benzo[c]carbazole core to develop asymmetric SAMs CbzBF and CbzBT, improving molecular packing, ITO work function modulation, and buried-interface passivation. However, an intrinsic incompat-

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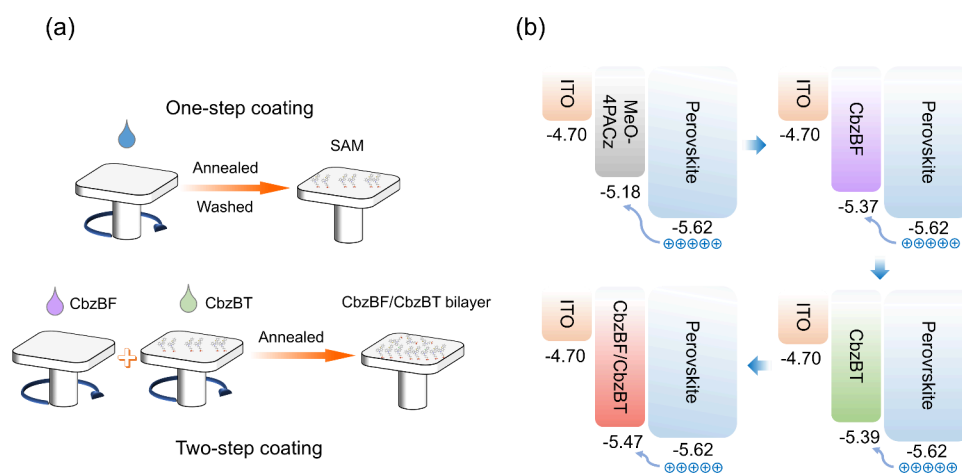


Figure 1. (a) Schematic diagram of sequential deposition steps. (b) HOMO energy-level arrangement of perovskite and its bottom interface functional layer.

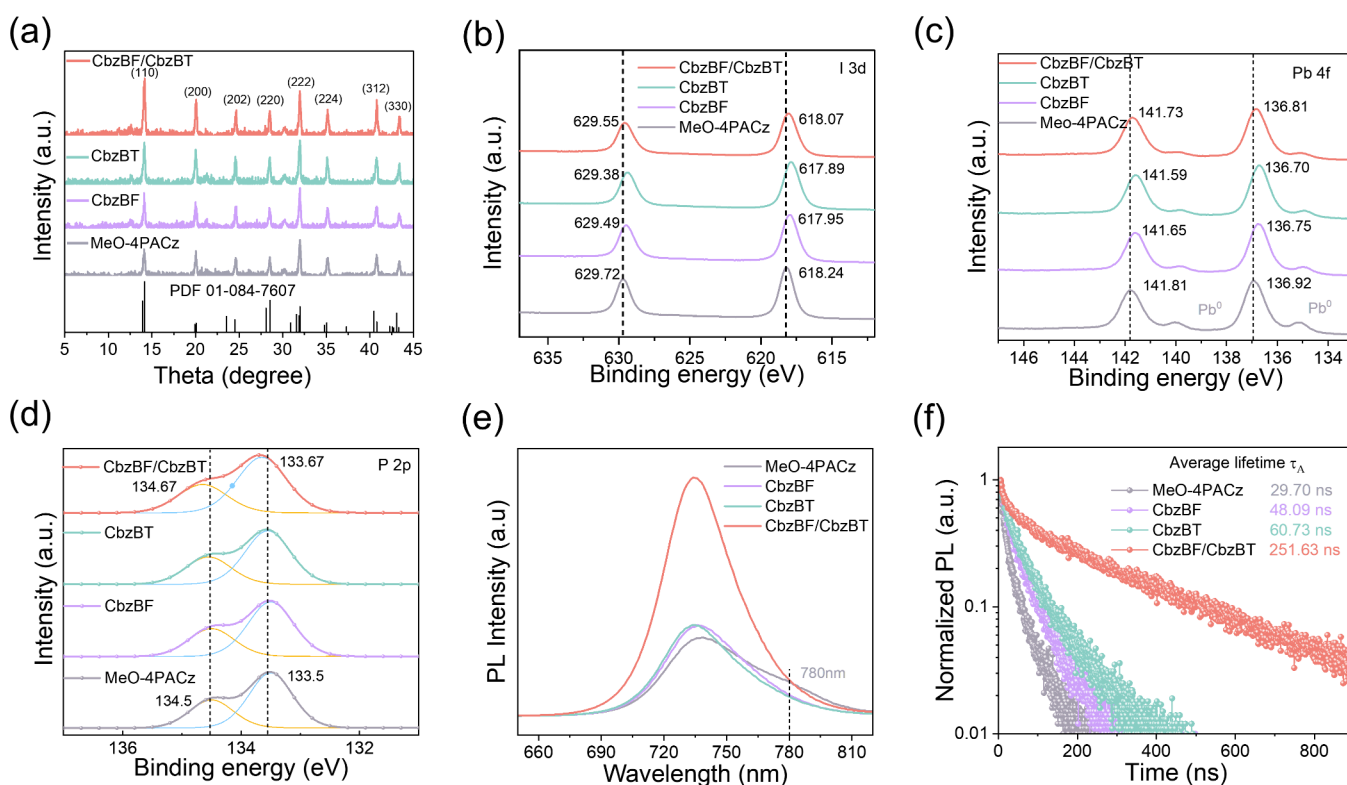


Figure 2. (a) XRD images of perovskite deposited on four types of HTLs. (b) I 3d 2P XPS spectra of perovskite deposited on four types of HTLs. (c) Pb 4f 2P XPS spectra of perovskite deposited on four types of HTLs. (d) P 2p XPS spectra of perovskite films fabricated from a 10-fold diluted precursor solution and deposited on four types of HTLs. (e) PL spectra of perovskite deposited on four types of HTLs. (f) TRPL spectra of perovskite deposited on four types of HTLs.

ibility exists within a single-SAM layer: achieving high surface coverage often requires molecular structures that provide limited chemical passivation at their terminals. Conversely, SAMs bearing highly functional passivating groups often suffer from poor assembly kinetics, resulting in patchy, pinhole-ridden monolayers on ITO.²² It is therefore highly desirable yet challenging to devise a novel approach that can integrate complete substrate coverage with effective terminal passivation, circumventing this fundamental constraint.

To address this, we propose an asymmetric self-assembled bilayer strategy. The bottom CbzBF SAM anchors the ITO substrate and optimizes energy-level alignment for efficient

hole extraction and high V_{OC} , while the top CbzBT SAM passivates perovskite defects and modulates crystallization, suppressing phase segregation and charge recombination. With this approach, 1.68 eV single-junction PPVs achieved a PCE of 22.86%, with a V_{OC} of 1.22 V, and a high fill factor (FF) of 86.54%. Under 1000 lx illumination from a 3000 K LED light source, the 1.68 eV PPVs attain a PCE of 43.88%, with a V_{OC} of 1.07 V, and a FF of 84%. Meanwhile, the universality of the CbzBF/CbzBT bilayer is demonstrated by high FFs (>85%) across perovskites with bandgaps of 1.65, 1.72, and 1.75 eV. These FF values rank among the highest reported in the literature. Furthermore, our two-terminal perovskite/silicon

tandem solar cells reach a PCE of 30.69%, with a V_{OC} of 1.94 V, and a FF of 82.04%.

RESULTS AND DISCUSSION

To systematically evaluate the device performance, the conventional symmetric SAM MeO-4PACz served as the benchmark against three hole-transport layer configurations: two asymmetric SAMs (CbzBF, CbzBT) and an asymmetric self-assembled bilayer (CbzBF/CbzBT). The molecular structures of MeO-4PACz, CbzBF, and CbzBT are shown in Figure S1. In contrast to the conventional one-step SAM deposition, our approach employs a two-step, sequential process to construct the CbzBF/CbzBT bilayer (Figure 1a), which involves first anchoring CbzBF to the substrate, followed by the assembly of CbzBT. X-ray photoelectron spectroscopy (XPS) measurements of S 2p peaks (Figure S2) confirmed that both CbzBF and CbzBT form robust interfaces on the substrate and remain intact after subsequent wet-processing steps. We further investigated the energy levels of MeO-4PACz, CbzBF, CbzBT, and the CbzBF/CbzBT bilayer using ultraviolet photoelectron spectroscopy (UPS). The measured HOMO levels for ITO/MeO-4PACz, ITO/CbzBF, ITO/CbzBT, and ITO/(CbzBF/CbzBT) were determined to be -5.18 , -5.37 , -5.39 , and -5.47 eV, respectively (Figure S3). Additionally, the HOMO level of the 1.68 eV perovskite was determined to be -5.62 eV from the UPS cutoff and onset regions (Figure S4). Based on these results, we established a comprehensive energy-level diagram of the ITO/HTL/perovskite structure (Figure 1b). The results indicate that the sequentially deposited CbzBF/CbzBT bilayer effectively modulates the interfacial energy alignment, thereby facilitating efficient charge extraction and transport.

To probe the influence of the CbzBF/CbzBT bilayer on perovskite crystallization, we first performed water contact angle measurements on various hole-transport layers (Figure S5). The contact angles for MeO-4PACz, CbzBF, and CbzBT were measured at 53.7° , 69.7° , and 71.4° , respectively, revealing that the asymmetric molecular structures enhance surface hydrophobicity. Remarkably, the sequentially deposited CbzBF/CbzBT bilayer exhibited a further increased contact angle of 75.5° . Surface wettability, often indicated by the water contact angle, is a key parameter affecting the nucleation density and crystal growth of perovskite films, as it governs the interaction between the precursor solution and the substrate. A moderately hydrophobic surface is generally correlated with a lower nucleation density, which can promote the formation of larger grains.^{23–25} This observation aligns with recent studies underscoring the significant role of interfacial properties in perovskite crystallization kinetics and final film morphology.²⁶

The progressively increased contact angles observed on our CbzBF, CbzBT, and CbzBF/CbzBT bilayer correspond well with the enhanced crystallinity and enlarged grain size demonstrated in our scanning electron microscopy (SEM) and X-ray diffraction (XRD) analyze.

The SEM images (Figure S6) and corresponding grain size analysis (Figure S7) showed that the average grain sizes of perovskite films grown on MeO-4PACz, CbzBF, CbzBT, and CbzBF/CbzBT were 207.4 ± 10 , 238.6 ± 11.8 , 275.9 ± 73.3 , and 277.3 ± 20.9 nm, respectively. Consistently, the XRD patterns (Figure 2a) demonstrated a gradual enhancement in the intensity of the characteristic (110) diffraction peak in the sequence of MeO-4PACz, CbzBF, CbzBT, and CbzBF/CbzBT. Together, the SEM and XRD results confirm that

the CbzBF/CbzBT bilayer effectively promotes the crystallization of perovskite films.

To gain deeper insight into the molecular-level interaction between the CbzBF/CbzBT bilayer and the perovskite, we performed X-ray photoelectron spectroscopy (XPS). Following previously reported methods, we extracted the peak positions (Tables S1–S2).^{27–29} As shown in Figure 2b,c, the I 3d and Pb 4f core-level peaks for perovskite films deposited on CbzBT, CbzBF, and the CbzBF/CbzBT bilayer all shifted to lower binding energies compared to the MeO-4PACz reference. This systematic shift suggests enhanced chemical interaction, which we attribute to the strong Lewis basicity of the oxygen and sulfur atoms in CbzBF and CbzBT. These atoms can donate lone pair electrons to undercoordinated Pb^{2+} , thereby passivating defects.^{30,31} Notably, the binding energy of the perovskite on the CbzBF/CbzBT bilayer did not shift toward a lower binding energy; instead, it exhibited a shift toward a higher binding energy compared to that on either CbzBF or CbzBT alone. Furthermore, as seen in Figure 2d, only the bilayer sample induced a positive shift in the P 2p peaks relative to their conventional positions (133.5 and 134.4 eV), indicating that the phosphate groups underwent a specific chemical reaction with the perovskite. This phosphate interaction further passivates the interface but introduces an electron-withdrawing effect, which increases the local binding energy of the adjacent Pb and I cores.^{32,33} Therefore, the final binding energy in the CbzBF/CbzBT bilayer is a result of these competing interactions, explaining why it does not show a further reduction compared to SAMs. Based on these observations, we conclude that the CbzBT layer in the bilayer structure provides dual-site passivation: its heteroatoms (O/S) coordinate with Pb^{2+} , while its phosphate groups chemically bond with the perovskite, synergistically reducing interface defects.

To probe the effects of perovskite-HTL interactions on charge transport and recombination dynamics, we conducted photoluminescence (PL) measurements on perovskite films deposited on the four SAMs. As shown in Figure 2e, a weak additional PL emission peak appears at around 780 nm for the film on MeO-4PACz, indicating the occurrence of phase separation.³⁴ In contrast, perovskites fabricated on CbzBT, CbzBF, and the CbzBF/CbzBT bilayer exhibit no such secondary emission features, demonstrating that the asymmetric SAMs based on CbzBF and CbzBT effectively suppress phase segregation. Moreover, the perovskite film prepared on the CbzBF/CbzBT bilayer exhibited the highest absolute PL intensity, indicating effective defect passivation at the buried HTL/perovskite interface.^{35,36} The time-resolved photoluminescence (TRPL) measurements were performed to observe the dynamic recombination (Figure 2f). The TRPL decay curves were fitted using a biexponential function expressed as

$$f(t) = A_1 \exp\left(-\frac{t}{\tau_1}\right) + A_2 \exp\left(-\frac{t}{\tau_2}\right) + \gamma_0 \quad (1)$$

where τ_1 represents the fast decay component, τ_2 the slow decay component, and A_1 and A_2 are the corresponding amplitudes. τ_1 typically represents fast nonradiative recombination occurring at the interface, often due to carrier trapping at defect sites. In contrast, τ_2 corresponds to the slower recombination process in the bulk perovskite, which is limited by carrier diffusion.

The average lifetime (τ_{avg}) was calculated using the formula:

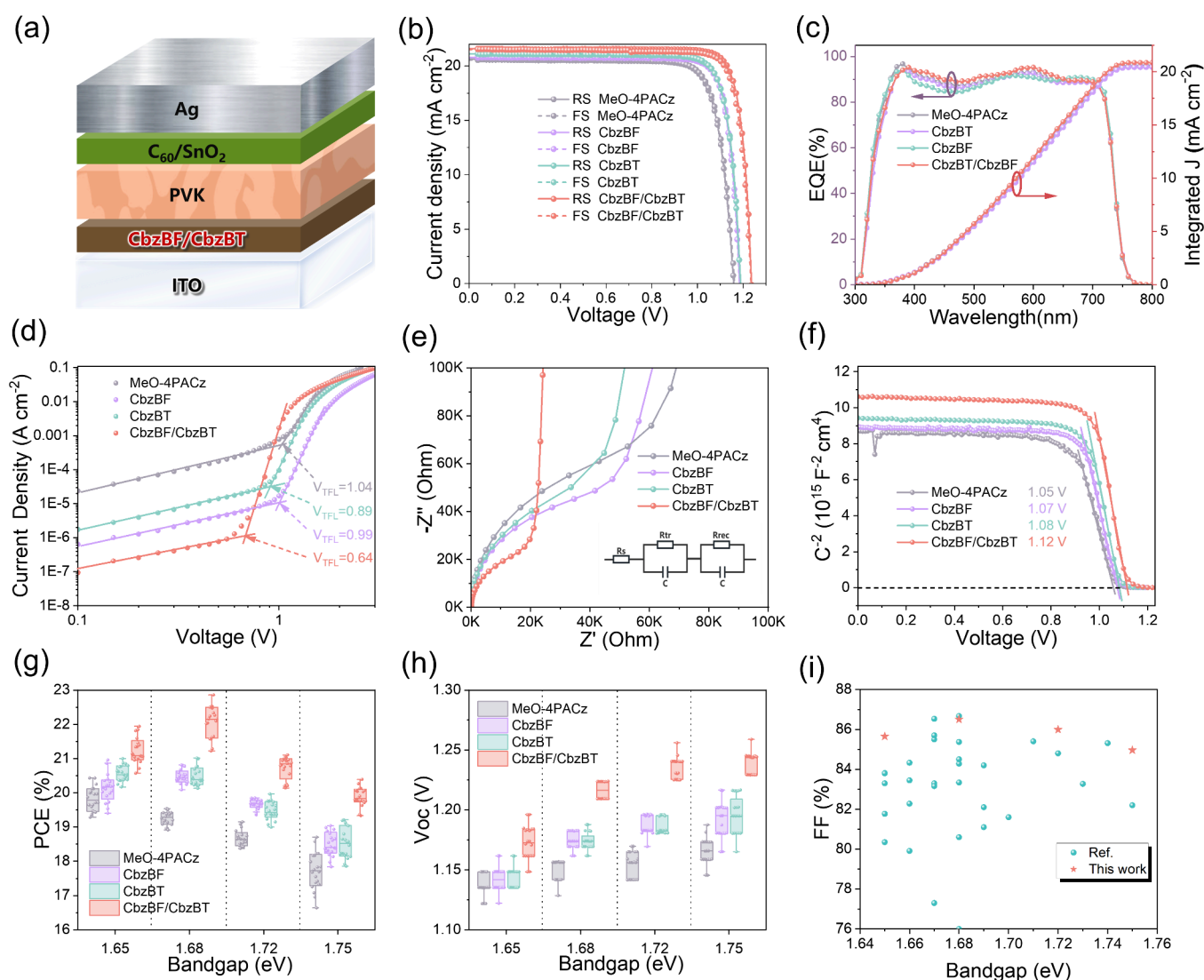


Figure 3. (a) Schematic diagram of the device structure. (b) J – V characteristics of PPVs devices obtained from four types of HTLs. (c) EQE curves of the champion devices obtained from four types of HTLs. (d) SCLC of the hole-only devices obtained from four types of HTLs. (e) Nyquist plot. (f) Capacitance–voltage curves of the devices obtained from four types of HTLs. (g, h) Statistical comparison of photovoltaic parameters for PPVs with different bandgaps (1.65, 1.68, 1.72, and 1.75 eV) employing various hole selective layers: (g) PCE and (h) FF. (i) FF values reported in the literature for wide-bandgap (WBG, 1.65–1.75 eV) PPVs. (Detailed data are provided in Table S8).

$$\tau_{\text{avg}} = \frac{A_1\tau_1^2 + A_2\tau_2^2}{A_1\tau_1 + A_2\tau_2} \quad (2)$$

As summarized in Table S3, the fitting results show that τ_{avg} of perovskite films prepared on MeO-4PACz, CbzBF, CbzBT, and CbzBF/CbzBT are 29.70, 48.09, 60.73, and 251.63 ns, respectively. The remarkably extended τ_{avg} observed for the CbzBF/CbzBT bilayer, compared to SAMs, stems from its superior capability in mitigating interfacial nonradiative recombination, as will be discussed later. This enhancement can be mechanistically understood by considering the synergistic functions of the sequentially deposited bilayer structure.

First, the bottom CbzBF layer provides robust anchoring to the ITO substrate, creating a foundation with optimized energy-level alignment for efficient hole extraction, as shown in Figure 1b and UPS data in Figure S3. This reduces interfacial energy barriers and associated recombination losses. Second, and more critically, the top CbzBT layer plays a dual role: (i)

Its heteroatoms (O and S) act as Lewis bases, donating lone pair electrons to coordinate with undercoordinated Pb^{2+} ions at the buried interface, effectively passivating these deep-level defects. (ii) Its molecular structure promotes favorable molecular stacking with the underlying CbzBF, leading to a more compact and ordered interfacial monolayer. This improved ordering minimizes pinholes and ensures uniform coverage, reducing shunt paths and areas prone to recombination.

To investigate the influence of different HTLs on device performance, we fabricated the wide-bandgap (1.68 eV) PPVs with the structure ITO/HTL/perovskite/ $\text{C}_{60}/\text{SnO}_2/\text{Ag}$ (Figure 3a). Figure 3b shows the current density–voltage (J – V) curves of devices with the corresponding photovoltaic parameters of the champion devices (Table S4). The results indicate that devices employing CbzBF, CbzBT, and CbzBF/CbzBT all exhibit significantly improved photovoltaic performance compared to the MeO-4PACz-based devices. The champion device is prepared with CbzBF/CbzBT bilayer,

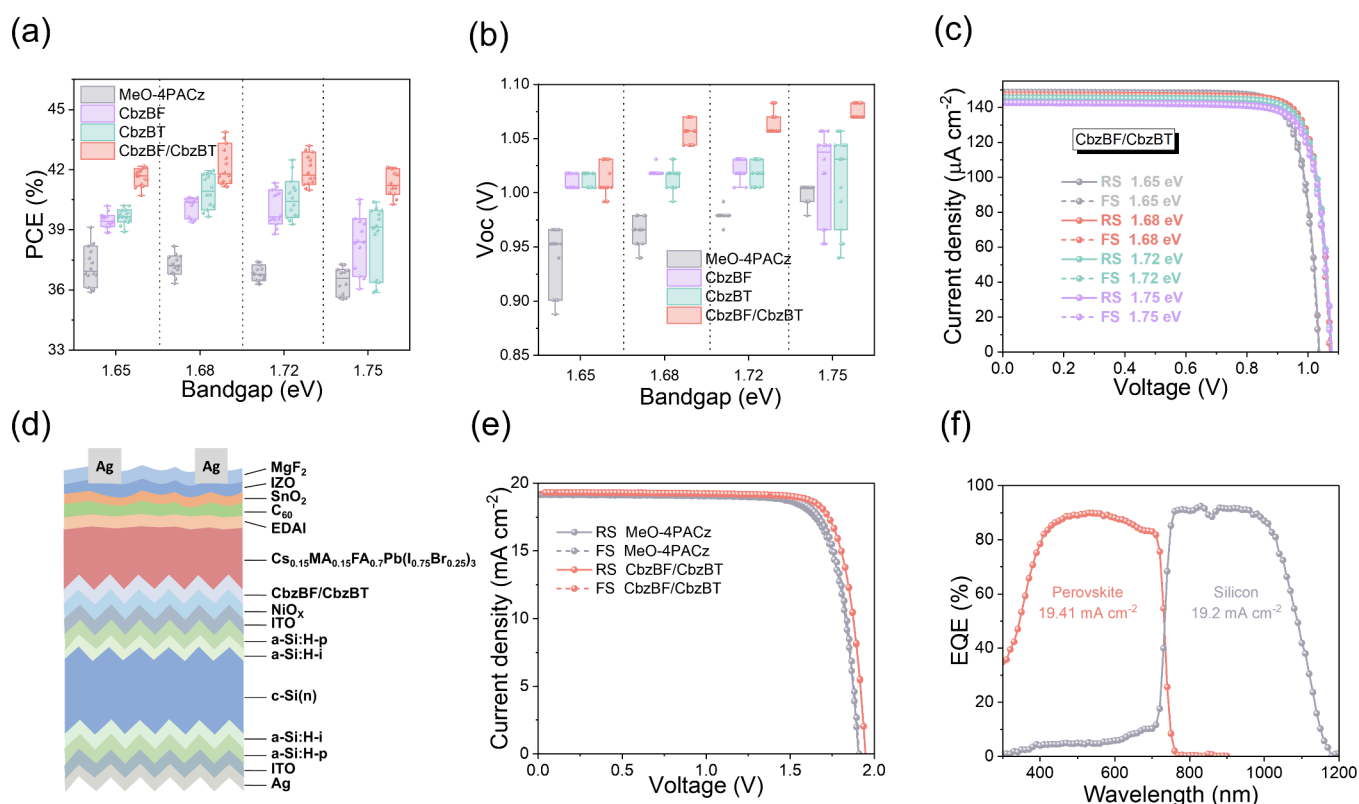


Figure 4. (a) PCE and (b) V_{OC} distributions for PPVs with bandgaps of 1.65, 1.68, 1.72, and 1.75 eV employing four types of HTLs, measured under 1000 lx illumination from a 3000 K LED light source. (c) J – V characteristics of PPVs with bandgaps of 1.65, 1.68, 1.72, and 1.75 eV using CbzBF/CbzBT as the hole transport layer, measured under 1000 lx illumination from a 3000 K LED light source. (d) Device architecture of the two-terminal perovskite/silicon tandem solar cell. (e) J – V characteristics of PPVs based on the two-terminal perovskite/silicon tandem solar cell. (f) EQE curves of the best-performing two-terminal perovskite/silicon tandem solar cell using CbzBF/CbzBT as the hole transport layer.

achieving a PCE of 22.86%, a significantly enhanced V_{OC} of 1.22 V (1.16 V for MeO-4PACz-based perovskite device), a J_{SC} of 21.58 $mA cm^{-2}$, and a FF of 86.54%. The external quantum efficiency (EQE) spectra of the champion devices with the four HTLs are presented in Figure 3c. All the integrated current densities are consistent with the J – V measurement results.

To evaluate the defect state density and charge carrier recombination behavior at the perovskite/HTL interface, we conducted space charge limited current (SCLC) measurements on hole-only devices (ITO/HTLs/perovskite/PTAA/Ag) (Figure 3e). The transition point between the ohmic region and the trap-filling region in the current–voltage curve corresponds to the trap-filling limit voltage (V_{TFL}).²⁵ The V_{TFL} values for devices based on MeO-4PACz, CbzBF, CbzBT, and CbzBF/CbzBT were 1.04, 0.99, 0.89, and 0.64 V, respectively. The defect state density (N_t) can be calculated using the following equation:

$$N_t = \frac{2\epsilon\epsilon_0 V_{TFL}}{ed^2} \quad (3)$$

where ϵ is the relative permittivity of the perovskite, ϵ_0 is the vacuum permittivity, e is the elementary charge, and d is the thickness of the perovskite layer. A lower V_{TFL} indicates a reduced trap state density, as N_t is positively correlated with V_{TFL} . The device with CbzBF/CbzBT exhibited the lowest V_{TFL} , confirming the lowest defect density in devices.

To further probe the influence of different HTLs on interface charge dynamics, electrochemical impedance spectroscopy (EIS) was performed. Figure 3f shows the Nyquist

plots of representative devices. The larger arc in the high-frequency region corresponds to the charge transfer process (charge transfer resistance, R_{tr}), while the smaller semicircle in the low-frequency region is associated with recombination behavior (recombination resistance, R_{rec}).^{37,38} Compared to the MeO-4PACz-based device, CbzBF-, CbzBT-, and CbzBF/CbzBT-based devices showed sequentially decreased R_{tr} and increased R_{rec} . The smallest R_{tr} and largest R_{rec} indicate that the CbzBF/CbzBT-based device effectively reduces nonradiative recombination and minimizes defect-assisted charge loss. To elucidate the mechanism behind the enhanced Voc in devices with CbzBF/CbzBT bilayer, capacitance–voltage (C – V) analysis was carried out (Figure 3g). The built-in potential (V_{bi}) values for devices based on MeO-4PACz, CbzBF, CbzBT, and CbzBF/CbzBT were measured to be 1.05, 1.07, 1.08, and 1.12 V, respectively. A higher V_{bi} indicates a stronger built-in electric field, which facilitates the separation and extraction of photogenerated carriers.³⁹

To validate the universality of CbzBF/CbzBT bilayer as a hole transport layer, we conducted a comprehensive performance comparison with MeO-4PACz, CbzBF, and CbzBT in PPVs with bandgaps of 1.65, 1.72, and 1.75 eV (Figure S8). Figures S9 and S10 indicate best ratio of CbzBF/CbzBT bilayer is 1:1. The J – V characteristics (Figure S11), EQE spectra (Figure S12), and champion device parameters (Tables S5–S7) consistently demonstrate that the CbzBF/CbzBT bilayer structure delivers optimal performance across all bandgap systems. Notably, high FFs of 85.65, 85.99, and 85.03% are achieved in 1.65, 1.72, and 1.75 eV perovskite,

respectively. Furthermore, we fabricated 20 independent devices for each bandgap (1.65, 1.68, 1.72, and 1.75 eV) and performed systematic statistical analysis of J_{SC} , V_{OC} , FF, and PCE (Figures 3g,h and S13). The results consistently confirm that the sequential deposition strategy of CbzBF/CbzBT markedly enhances device performance and reproducibility. Finally, by comparing the FF values obtained in this work with recently reported data for 1.65–1.75 eV single-junction perovskite cells (Figure 3i), we observed that devices with CbzBF/CbzBT bilayer achieve high FFs within the highest reported range in the literature. These findings robustly attest to the technical advancement and broad application potential of the CbzBF/CbzBT bilayer strategy for developing highly efficient and stable PPVs.

Then, we evaluate the performance of wide-bandgap PPVs for indoor photovoltaic (LED, 1000 lx, $301.9 \mu\text{W}\cdot\text{cm}^{-2}$) applications. As shown in Figures 4a,b and S14, compared to the MeO-4PACz-based PPVs, CbzBF, CbzBT, and the CbzBF/CbzBT bilayer HTL deliver significant improvements in both V_{OC} and J_{SC} , leading to a notable enhancement in average PCE. The J – V characteristics and champion device parameters are summarized in Figure 4e and Table S9, respectively. Notably, PCEs of 41.92, 43.88, 43.19, and 42.12% were achieved in perovskite devices with bandgaps of 1.65, 1.68, 1.72, and 1.75 eV. Corresponding high V_{OC} of 1.031, 1.07, 1.083, and 1.083 V were also obtained, further affirming the superior performance enabled by the sequential deposition of the CbzBF/CbzBT bilayer. Furthermore, the sequential deposition method for the CbzBF/CbzBT bilayer was extended to two-terminal perovskite/silicon tandem solar cells. The tandem device featured a silicon bottom cell with an asymmetrical pyramid texture (small on the front, large on the rear), paired with a 1.68 eV perovskite top cell (Figure 4d). The champion tandem device incorporating CbzBF/CbzBT bilayer as the HTL achieves a power conversion efficiency of 30.69%, with a V_{OC} of 1.94 V, J_{SC} of $19.32 \text{ mA}/\text{cm}^2$, and a FF of 82.04% (Figure 4e and Table S10). Moreover, the EQE analysis (Figure 4f) demonstrated closely matched integrated current densities between the subcells, confirming successful current matching in the tandem architecture.

Although experimentally validated here for the 1.65–1.75 eV range crucial for tandem cells, the fundamental advantages of this bilayer design are not inherently bandgap-limited. Thus, this strategy offers a promising platform adaptable to perovskites with differing bandgaps, such as those below 1.6 eV or above 1.8 eV, meriting further exploration. The superior charge-collection efficiency (FF > 85%) also implies strong potential for stable performance under the varied light intensities of real-world indoor environments (e.g., 200–500 lx), which is key for practical applications such as Internet-of-Things devices.

CONCLUSIONS

In summary, this work demonstrates that a sequentially deposited self-assembled bilayer CbzBF/CbzBT significantly enhances the performance of wide-bandgap PPVs across single-junction, indoor, and tandem architectures. By strategically engineering the HTL/perovskite interface with this bilayer design, we achieved remarkable improvements in key photovoltaic parameters, particularly V_{OC} and FF. Notably, WBG devices with bandgaps from 1.65 to 1.75 eV consistently achieved high FFs, all exceeding 85% and ranking among the highest reported values. Furthermore, the bilayer enabled a

champion PCE of 43.88% under indoor LED illumination, along with a 30.69% efficient two-terminal perovskite/silicon tandem cell. These results underscore the broad applicability and scalability of our interfacial strategy. This study provides not only an effective approach for stabilizing and optimizing buried interfaces but also a generalizable design principle for future high-performance perovskite photovoltaics. More broadly, the sequential-deposition bilayer strategy holds great promise for extension to other optoelectronic devices.

EXPERIMENTAL SECTION

Materials

Formamidinium iodide (FAI, $\geq 99.5\%$) was purchased from Greatcell Solar, Australia. Methylammonium bromide (MABr , $\geq 99.995\%$), lead iodide (PbI_2 , $\geq 99.99\%$), lead bromide (PbBr_2 , $\geq 99.99\%$) and cesium iodide (CsI , $\geq 99.999\%$), [4-(3,6-Dimethoxy-9H-carbazol-9-yl)butyl]phosphonic acid (MeO-4PACz), Ethane-1,2-diaminium Iodide (EDAI_2), and Poly[bis(4-phenyl)(2,4,6-trimethylphenyl)-amine] (PTAA) were purchased from Xian e-Light New Material Company limited. C_{60} was purchased from Xian Polymer Light Technology Corp. *N,N*-dimethylformamide (DMF, $\geq 99.8\%$), dimethyl sulfoxide (DMSO, $\geq 99.9\%$), isopropanol (IPA, $\geq 99.9\%$), ethyl acetate ($\geq 99\%$), Toluene ($\geq 99.9\%$) was purchased from Guangzhou Chemical Reagent Factory. All the reagents were used as received without further purification. CbzBT and CbzBF were obtained from the City University of Hong Kong.

Preparation of Perovskite Precursor

Perovskite precursor solutions with targeted bandgaps of 1.65, 1.68, 1.72, and 1.75 eV were synthesized using a consistent formulation approach, wherein the bandgap was precisely modulated by systematically adjusting the bromide-to-iodide ratio in the lead halide composition. The corresponding chemical formulations and reagent quantities were as follows: the 1.65 eV composition $\text{Cs}_{0.15}\text{FA}_{0.70}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$ contained 21.84 mg MABr , 50.66 mg CsI , 156.52 mg FAI , 118.91 mg PbBr_2 , and 470.95 mg PbI_2 ; the 1.68 eV analogue $\text{Cs}_{0.15}\text{FA}_{0.70}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.75}\text{Br}_{0.25})_3$ used the same organic and cesium salt quantities with 143.13 mg PbBr_2 and 425.97 mg PbI_2 ; the 1.72 eV variant $\text{Cs}_{0.15}\text{FA}_{0.70}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.7}\text{Br}_{0.3})_3$ maintained identical organic and cesium precursor masses while incorporating 178.92 mg PbBr_2 and 380.21 mg PbI_2 ; and the 1.75 eV composition $\text{Cs}_{0.15}\text{FA}_{0.70}\text{MA}_{0.15}\text{Pb}(\text{I}_{0.65}\text{Br}_{0.35})_3$ employed 214.70 mg PbBr_2 and 329.62 mg PbI_2 alongside consistent MABr , CsI , and FAI amounts. All compounds were dissolved in a mixed solvent of DMF and DMSO (4:1, v/v), stirred at 60 °C for 2 h, cooled to room temperature, and filtered through a 0.22 μm PTFE syringe filter before use.

Fabrication of Wide-Bandgap Perovskite Solar Cells

The ITO glass substrates were cleaned through sequential ultrasonic treatment in a diluted glass cleaning agent, deionized water, and alcohol, each for 15 min. Subsequently, the substrates were subjected to UV-ozone treatment for 10 min. SAM solutions of MeO-4PACz, CbzBF, and CbzBT were prepared by dissolving each material in a mixed solvent of IPA/DMF (100/1, v/v) at a concentration of 0.5 mg/mL under stirring at 60 °C for 2 h and subsequently cooled to room temperature before use. For the sequential deposition of the CbzBF/CbzBT bilayer, solutions of CbzBF and CbzBT were prepared separately at a concentration of 0.25 mg mL^{-1} using the same solvent mixture and preparation protocol. For the deposition of MeO-4PACz HTL, 120 μL of a 0.5 mg mL^{-1} solution was spin-coated onto the ITO substrate at 3000 rpm for 30 s, followed by annealing at 100 °C for 10 min. The substrate was then washed with IPA (150 μL) via spin-coating at 3000 rpm for 30 s and annealed again at 100 °C for 5 min. The same procedure was applied for CbzBF and CbzBT HTLs, using their respective 0.5 mg mL^{-1} solutions in place of MeO-4PACz. For deposition of CbzBF/CbzBT HTL, 120 μL of 0.25 mg mL^{-1} CbzBF was added to the ITO and spin-coated at 3000 rpm for 30 s, followed by annealing at 100 °C for 10 min. 120 μL of 0.25 mg mL^{-1}

CbzBT was added to the ITO and spin-coated at 3000 rpm for 30 s, followed by annealing at 100 °C for 10 min. The perovskite films were deposited on the substrates using a third-step spin coating procedure. The first step involved spinning at 1000 rpm for 3 s with an acceleration of 350 rpm s⁻¹. The third step involved spinning at 4000 rpm for 35 s with an acceleration of 700 rpm s⁻¹. During the third spin-coating step, 250 μ L of EA was dropped onto the spinning substrate at 25 s before the end of the procedure. The substrates were then transferred to a hot plate and heated using a two-step procedure: 60 °C for 2 min, followed by 100 °C for 15 min. After cooling to room temperature, 150 μ L solution of EDAL₂ in IPA (0.7 mg mL⁻¹) was dropped onto the perovskite film and spin-coated at 4000 rpm for 30 s. The substrate was then annealed at 100 °C for 10 min. The 20 nm C₆₀ layer was deposited by thermal evaporation with an evaporation rate of 0.3–0.5 Å/s. The 20 nm SnO₂ layer was deposited by atomic layer deposition (ALD) in a reactor. TDMASn was used as the Sn precursor and held at 60 °C in a stainless-steel container. Water was used as an oxidant and delivered from a stainless-steel container without active heating, whereas the precursor delivery manifold and deposition reactor were heated to 110 °C. Each ALD cycle consisted of an H₂O vapor dose of 0.2 s, followed by a purge of 3.0 s, and then a TDMASn dose of 0.06 s, followed by a purge of 3.0 s. With this, 200 cycles led to 20 nm SnO₂. Finally, 150 nm silver (Ag) contacts were thermally evaporated on the active area (0.09 cm²).

Fabrication of Two-Terminal Perovskite/Silicon Tandem Solar Cells

The Czochralski n-Si wafer types were first ozone-cleaned, then dipped in 1% HF to generate the desired texture. Four nm i-a-Si: H/4 nm n⁺-a-Si: H electron transport layer and 6 nm i-a-Si: H/13 nm p⁺-a-Si: H hole transport layer were added. Transparent electrodes followed—20 nm ITO on the n-side and 70 nm ITO on the p-side—by DC magnetron sputtering in Ar/O₂. 1000 nm-Silver fingers were thermally evaporated onto the p-ITO. Before coating the top perovskite solar cells, the SHJ bottom cells were annealed in ambient air (200 °C, 20 min) and given a 5 min UV-ozone bath. Perovskite solar cells with a nickel oxide (NiO_x) hole-transport layer, which was deposited by RF magnetron sputtering at 65 W, 2 Pa for 20 min, were spun on the n-side of the silicon heterojunction. The precursor concentration was raised from 1.3 to 1.4 M relative to single-junction references. Electron-transport C₆₀ was thinned to 12 nm, and ALD-SnO₂ was cut to 140 cycles. A 50 nm IZO semitransparent top contact was RF-sputtered (30 W, 0.4 Pa) and capped with 200 nm Ag. Using a 115 nm thermally evaporated MgF₂ foil to reduce the reflection. The photoactive area was set with a 1 cm² aperture mask.

Device Characterizations

The current density–voltage (*J*–*V*) characteristics were characterized under standard AM 1.5G illumination (100 mW·cm⁻²) using a Newport solar simulator (Oriel SOI3A). Capacitance–voltage (*C*²–*V*) profiling was carried out to determine the net carrier concentration (*N*_{CV}). The *C*²–*V* measurements were performed under dark conditions at room temperature using an AC frequency of 15 kHz and a modulation amplitude of 20 mV. The space-charge-limited current (SCLC) characteristics of the perovskite devices were evaluated using a commercial measurement system (XPQY-eq E-350-1100, Guangzhou Xipu Optoelectronics Technology) equipped with an integrating sphere (Labsphere GPS-4P-SL) and a photo-detector array (Hamamatsu S7031-1006). External quantum efficiency (EQE) spectra were obtained using an Enlitech QE-R system calibrated with a certified silicon reference cell. For two-terminal perovskite/silicon tandem solar cells, the EQE of the silicon bottom subcell was measured using a 500 nm short-pass optical filter, while that of the perovskite top subcell was measured using an 850 nm long-pass filter. Static contact angles were measured with a high-resolution camera fitted with drop shape analysis software. Morphological characterization was conducted using field-emission scanning electron microscopy (FE-SEM; FEI Apreo LoVac). Ultraviolet photoelectron spectroscopy (UPS) was performed on a

Thermo Fisher Scientific ESCALAB 250 system with a He I excitation source (21.22 eV). Photoluminescence (PL) and time-resolved photoluminescence (TRPL) spectra were acquired using an Edinburgh Instruments FLS980 spectrofluorometer. X-ray Photoelectron Spectroscopy (XPS) measurements were conducted on the 10-fold diluted perovskite thin films using a Nexsa spectrometer (Thermo Fisher Scientific) with a monochromatic Al K α X-ray source under an ultrahigh vacuum of 2 $\times$ 10⁻⁷ Pa. X-ray diffraction (XRD) patterns of the perovskite films were recorded using a Bruker D8 Advance diffractometer with Cu K α radiation operated at 40 kV and 40 mA.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c24604>.

Molecular structures of MeO-4PACz, CbzBF, and CbzBT; XPS spectra of S 2P in the perovskite films deposited on the four types of HTLs; UPS spectra for energy-level alignment analysis of the four HTLs and the perovskite; water contact angles of four different HTLs; SEM images of perovskite films fabricated on four different HTLs; grain size distribution histograms of perovskite films deposited on four different HTLs; EQE spectra and derived bandgaps of the champion devices; statistics on process-optimized PPVs; *J*–*V* curves of the champion single-junction PSCs with different bandgap; photovoltaic performance parameters (PCE, *V*_{OC}, *J*_{SC}, and FF) for devices with different bandgap; TRPL fitting parameters of perovskite; information of representative WBG PSCs; and photovoltaic performance parameters of the champion two-terminal perovskite/silicon tandem solar cell (PDF)

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Author Contributions

Y.W. and C.H. conceived the idea. C.H. and J.Z. performed experiments and analysis. C.H., J.Z., and Y.W. wrote the manuscript. Y.L. and Y.W. contributed to the manuscript review and editing. Y.W. supervised this project. All authors discussed the results and commented on the manuscript. C.H. and J.Z. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) An, Y.; Perini, C. A. R.; Hidalgo, J.; Castro-Mendez, F.; Vagott, J. N.; Li, R.; Saidi, W. A.; Wang, S.; Li, X.; Correa-Baena, J.-P. Identifying High-Performance and Durable Methylammonium-Free Lead Halide Perovskites via High-Throughput Synthesis and Characterization. *Energy Environ. Sci.* **2021**, *14* (12), 6638–6654.
- (2) Xiao, T.; Tu, S.; Liang, S.; Guo, R.; Tian, T.; Müller-Buschbaum, P. Solar Cell-Based Hybrid Energy Harvesters towards Sustainability. *Opto-Electron. Sci.* **2023**, *2* (6), No. 230011.
- (3) Kojima, A.; Teshima, K.; Shirai, Y.; Miyasaka, T. Organometal Halide Perovskites as Visible-Light Sensitizers for Photovoltaic Cells. *J. Am. Chem. Soc.* **2009**, *131* (17), 6050–6051.
- (4) Cai, M.; Wu, Y.; Chen, H.; Yang, X.; Qiang, Y.; Han, L. Cost-Performance Analysis of Perovskite Solar Modules. *Adv. Sci.* **2017**, *4* (1), No. 1600269.
- (5) Yang, Y.; Yang, M.; Li, Z.; Crisp, R.; Zhu, K.; Beard, M. C. Comparison of Recombination Dynamics in CH₃NH₃PbBr₃ and CH₃NH₃PbI₃ Perovskite Films: Influence of Exciton Binding Energy. *J. Phys. Chem. Lett.* **2015**, *6* (23), 4688–4692.
- (6) Shi, Z.-E.; Cheng, T.-H.; Lung, C.-Y.; Lin, C.-W.; Wang, C.-L.; Jiang, B.-H.; Hsiao, Y.-S.; Chen, C.-P. Achieving over 42% Indoor Efficiency in Wide-Bandgap Perovskite Solar Cells through Optimized Interfacial Passivation and Carrier Transport. *Chem. Eng. J.* **2024**, *498*, No. 155512.
- (7) Ma, Q.; Wang, Y.; Liu, L.; Yang, P.; He, W.; Zhang, X.; Zheng, J.; Ma, M.; Wan, M.; Yang, Y.; Zhang, C.; Mahmoudi, T.; Wu, S.; Liu, C.; Hahn, Y.-B.; Mai, Y. One-Step Dual-Additive Passivated Wide-Bandgap Perovskites to Realize 44.72%-Efficient Indoor Photovoltaics. *Energy Environ. Sci.* **2024**, *17* (5), 1637–1644.
- (8) Liu, B.; Jia, Y.; Yuan, C.; Wang, L.; Gao, X.; Yin, S.; Xu, J. Safety Issues and Mechanisms of Lithium-Ion Battery Cell upon Mechanical Abusive Loading: A Review. *Energy Storage Mater.* **2020**, *24*, 85–112.
- (9) Li, M.; Zhao, C.; Wang, Z.-K.; Zhang, C.-C.; Lee, H. K.; Pockett, A.; Barbé, J.; Tsoi, W. C.; Yang, Y.-G.; Carnie, M. J.; Gao, X.-Yu.; Yang, W.-X.; Durrant, J. R.; Liao, L.-S.; Jain, S. M. Interface Modification by Ionic Liquid: A Promising Candidate for Indoor Light Harvesting and Stability Improvement of Planar Perovskite Solar Cells. *Adv. Energy Mater.* **2018**, *8* (24), No. 1801509.
- (10) Leijtens, T.; Bush, K. A.; Prasanna, R.; McGehee, M. D. Opportunities and Challenges for Tandem Solar Cells Using Metal Halide Perovskite Semiconductors. *Nat. Energy* **2018**, *3* (10), 828–838.
- (11) Zheng, J.; Lau, C. F. J.; Mehrvarz, H.; Ma, F.-J.; Jiang, Y.; Deng, X.; Soeriyadi, A.; Kim, J.; Zhang, M.; Hu, L.; Cui, X.; Lee, D. S.; Bing, J.; Cho, Y.; Chen, C.; Green, M. A.; Huang, S. A.; Ho-Baillie, W. Y. Large Area Efficient Interface Layer Free Monolithic Perovskite/Homo-Junction-Silicon Tandem Solar Cell with over 20% Efficiency. *Energy Environ. Sci.* **2018**, *11* (9), 2432–2443.
- (12) Guo, Y.; Du, S.; Chen, W.; Zhou, H.; Chen, G.; Wang, S.; Yu, Z.; Hu, X.; Yao, F.; Li, C.; Ke, W.; Fang, G. Contact Potential Homogenization via Buried Interface Engineering Enables High-Performance Wide-Bandgap Perovskite Photovoltaics. *Adv. Funct. Mater.* **2025**, *35*, No. 2500168.
- (13) Liu, J.; He, Y.; Ding, L.; Zhang, H.; Li, Q.; Jia, L.; Yu, J.; Lau, T. W.; Li, M.; Qin, Y.; Gu, X.; Zhang, F.; Li, Q.; Yang, Y.; Zhao, S.; Wu, X.; Liu, J.; Liu, T.; Gao, Y.; Wang, Y.; Dong, X.; Chen, H.; Li, P.; Zhou, T.; Yang, M.; Ru, X.; Peng, F.; Yin, S.; Qu, M.; Zhao, D.; Zhao, Z.; Li, M.; Guo, P.; Yan, H.; Xiao, C.; Xiao, P.; Yin, J.; Zhang, X.; Li, Z.; He, B.; Xu, X. Perovskite/Silicon Tandem Solar Cells with Bilayer Interface Passivation. *Nature* **2024**, *635* (8039), 596–603.
- (14) Green, M. A.; Dunlop, E. D.; Yoshita, M.; Kopidakis, N.; Bothe, K.; Siefert, G.; Hao, X.; Jiang, J. Y. Solar Cell Efficiency Tables (Version 66). *Prog. Photovolt.* **2025**, *33*, 795–810.
- (15) Li, Y.; Yang, S.; Yan, S.; Liu, X.; Shi, G.; He, T. Open-Circuit Voltage Loss Management for Efficient Inverted Wide-Bandgap Perovskite Photovoltaics. *Adv. Funct. Mater.* **2025**, *35* (7), No. 2415331.
- (16) Zhang, X.; Wu, S.; Zhang, H.; Jen, A. K. Y.; Zhan, Y.; Chu, J. Advances in Inverted Perovskite Solar Cells. *Nat. Photonics* **2024**, *18* (12), 1243–1253.
- (17) Park, S. J.; Jeon, S.; Lee, I. K.; Zhang, J.; Jeong, H.; Park, J.-Y.; Bang, J.; Ahn, T. K.; Shin, H. W.; Kim, B. G.; Park, H. J. Inverted Planar Perovskite Solar Cells with Dopant Free Hole Transporting Material: Lewis Base-Assisted Passivation and Reduced Charge Recombination. *J. Mater. Chem. A* **2017**, *5* (25), 13220–13227.
- (18) Huang, C.; Fu, W.; Li, C.-Z.; Zhang, Z.; Qiu, W.; Shi, M.; Heremans, P.; Jen, A. K.-Y.; Chen, H. Dopant-Free Hole-Transporting Material with a C_{3h} Symmetrical Truxene Core for Highly Efficient Perovskite Solar Cells. *J. Am. Chem. Soc.* **2016**, *138* (8), 2528–2531.
- (19) Jiang, W.; Liu, M.; Li, Y.; Lin, F. R.; Jen, A. K.-Y. Rational Molecular Design of Multifunctional Self-Assembled Monolayers for Efficient Hole Selection and Buried Interface Passivation in Inverted Perovskite Solar Cells. *Chem. Sci.* **2024**, *15* (8), 2778–2785.
- (20) Wu, W.; Gao, H.; Jia, L.; Li, Y.; Zhang, D.; Zhan, H.; Xu, J.; Li, B.; Geng, Z.; Cheng, Y.; Tong, H.; Pan, Y.; Liu, J.; He, Y.; Xu, X.; Li, Z.; He, B.; Zhou, M.; Wang, L.; Qin, C. Stable and Uniform Self-Assembled Organic Diradical Molecules for Perovskite Photovoltaics. *Science* **2025**, *389* (6756), 195–199.
- (21) Park, S. J.; Yu, C.; Shim, K. I.; Hong, G. P.; Song, S.; Park, J. H.; Hwang, S. K.; Choi, J.; Han, J. W.; Kwon, M. S.; Park, I. J.; Kim, J. Y. Tailored Self-Assembled Monolayer Molecules for Perovskite/PERC Tandem Solar Cells with Efficiencies over 30%. *Energy Environ. Sci.* **2025**, *18*, 9105–9113.
- (22) Zhang, X.; Wang, Y.; Zhang, K.; Tao, M.; Guo, H.; Guo, L.; Song, Z.; Wen, J.; Yang, Y.; Hou, Y.; Song, Y. Reinforcing Coverage of Self-assembled Monomolecular Layers for Inverted Perovskite Solar Cells with Efficiency of 25.70%. *Angew. Chem., Int. Ed.* **2025**, *64* (13), No. e202423827.
- (23) Jiang, W.; Li, F.; Li, M.; Qi, F.; Lin, F. R.; Jen, A. K.-Y. π -Expanded Carbazoles as Hole-Selective Self-Assembled Monolayers

for High-Performance Perovskite Solar Cells. *Angew. Chem., Int. Ed.* **2022**, *134* (51), No. e202213560.

(24) Deng, Y.; Zheng, X.; Bai, Y.; Wang, Q.; Zhao, J.; Huang, J. Surfactant-Controlled Ink Drying Enables High-Speed Deposition of Perovskite Films for Efficient Photovoltaic Modules. *Nat. Energy* **2018**, *3* (7), 560–566.

(25) Liu, M.; Bi, L.; Jiang, W.; Zeng, Z.; Tsang, S.; Lin, F. R.; Jen, A. K.-Y. Compact Hole-Selective Self-Assembled Monolayers Enabled by Disassembling Micelles in Solution for Efficient Perovskite Solar Cells. *Adv. Mater.* **2023**, *35* (46), No. 2304415.

(26) Li, N.; Pratap, S.; Guo, R.; He, Z.; Liang, S.; Jia, X.; Gholipour, M.; Babbe, F.; Barchi, N. S.; Slack, J. L.; Tamura, N.; Qiao, L.; Sutter-Fella, C. M.; Müller-Buschbaum, P. Unveiling the Role of Halide Mixing in the Crystallization Kinetics and Charge Transfer Mechanisms of Wide-Bandgap Organic–Inorganic Halide Perovskites. *Energy Environ. Sci.* **2025**, *18*, 10460–10472.

(27) Guo, R.; Han, D.; Chen, W.; Dai, L.; Ji, K.; Xiong, Q.; Li, S.; Reb, L. K.; Scheel, M. A.; Pratap, S.; Li, N.; Yin, S.; Xiao, T.; Liang, S.; Oechsle, A. L.; Weindl, C. L.; Schwartzkopf, M.; Ebert, H.; Gao, P.; Wang, K.; Yuan, M.; Greenham, N. C.; Stranks, S. D.; Roth, S. V.; Friend, R. H.; Müller-Buschbaum, P. Degradation Mechanisms of Perovskite Solar Cells under Vacuum and One Atmosphere of Nitrogen. *Nat. Energy* **2021**, *6* (10), 977–986.

(28) Li, N.; Guo, R.; Chen, W.; Körstgens, V.; Heger, J. E.; Liang, S.; Brett, C. J.; Hossain, M. A.; Zheng, J.; Deimel, P. S.; Buyruk, A.; Allegretti, F.; Schwartzkopf, M.; Veinot, J. G. C.; Schmitz, G.; Barth, J. V.; Ameri, T.; Roth, S. V.; Müller-Buschbaum, P. Tailoring Ordered Mesoporous Titania Films via Introducing Germanium Nanocrystals for Enhanced Electron Transfer Photoanodes for Photovoltaic Applications. *Adv. Funct. Mater.* **2021**, *31* (34), No. 2102105.

(29) Liang, J.; Hu, X.; Wang, C.; Liang, C.; Chen, C.; Xiao, M.; Li, J.; Tao, C.; Xing, G.; Yu, R.; Ke, W.; Fang, G. Origins and Influences of Metallic Lead in Perovskite Solar Cells. *Joule* **2022**, *6* (4), 816–833.

(30) Fei, C.; Li, N.; Wang, M.; Wang, X.; Gu, H.; Chen, B.; Zhang, Z.; Ni, Z.; Jiao, H.; Xu, W.; Shi, Z.; Yan, Y.; Huang, J. Lead-Chelating Hole-Transport Layers for Efficient and Stable Perovskite Mini-modules. *Science* **2023**, *380* (6647), 823–829.

(31) Ji, X.; Bi, L.; Fu, Q.; Li, B.; Wang, J.; Jeong, S. Y.; Feng, K.; Ma, S.; Liao, Q.; Lin, F. R.; Woo, H. Y.; Jen, A. K.-Y.; Guo, X. Target Therapy for Buried Interface Enables Stable Perovskite Solar Cells with 25.05% Efficiency. *Adv. Mater.* **2023**, *35* (39), No. 2303665.

(32) Xu, P.; Hou, X.; Sun, X.; Zhang, J.; Zhang, W.; Zhao, X. Stepwise energy level regulation via bilayer self-assembled hole-transport materials for efficient and stable inverted perovskite solar cells. *J. Energy Chem.* **2025**, *109*, 8–14.

(33) Cui, H.; Huang, L.; Zhou, S.; Wang, C.; Hu, X.; Guan, H.; Wang, S.; Shao, W.; Pu, D.; Dong, K.; Zhou, J.; Jia, P.; Tao, C.; Ke, W.; Fang, G. Lead halide coordination competition at buried interfaces for low V_{OC}-deficits in wide-bandgap perovskite solar cells. *Energy Environ. Sci.* **2023**, *16* (12), 5992–6002.

(34) Wang, Z.; Zeng, L.; Zhu, T.; Chen, H.; Chen, B.; Kubicki, D. J.; Balvanz, A.; Li, C.; Maxwell, A.; Ugur, E.; Dos Reis, R.; Cheng, M.; Yang, G.; Subedi, B.; Luo, D.; Hu, J.; Wang, J.; Teale, S.; Mahesh, S.; Wang, S.; Hu, S.; Jung, E. D.; Wei, M.; Park, S. M.; Gräter, L.; Aydin, E.; Song, Z.; Podraza, N. J.; Lu, Z.-H.; Huang, J.; Dravid, V. P.; De Wolf, S.; Yan, Y.; Grätzel, M.; Kanatzidis, M. G.; Sargent, E. H. Suppressed Phase Segregation for Triple-Junction Perovskite Solar Cells. *Nature* **2023**, *618* (7963), 74–79.

(35) Stolterfoht, M.; Wolff, C. M.; Márquez, J. A.; Zhang, S.; Hages, C. J.; Rothhardt, D.; Albrecht, S.; Burn, P. L.; Meredith, P.; Unold, T.; Neher, D. Visualization and Suppression of Interfacial Recombination for High-Efficiency Large-Area Pin Perovskite Solar Cells. *Nat. Energy* **2018**, *3* (10), 847–854.

(36) Farag, A.; Feeney, T.; Hossain, I. M.; Schackmar, F.; Fassl, P.; Küster, K.; Bäuerle, R.; Ruiz-Preciado, M. A.; Hentschel, M.; Ritzer, D. B.; Diercks, A.; Li, Y.; Nejand, B. A.; Laufer, F.; Singh, R.; Starke, U.; Paetzold, U. W. Evaporated Self-Assembled Monolayer Hole

Transport Layers: Lossless Interfaces in P-i-n Perovskite Solar Cells. *Adv. Energy Mater.* **2023**, *13* (8), No. 2203982.

(37) Cao, X.; Qi, Y.; Zhang, J.; Hu, T.; Gao, Z.; Tang, B.; Zhang, S. Full-Cycle Humidity-Tolerant Perovskite Solar Cells Enabled by Air-Processed In Situ Shielding via Synergistic Dynamic Bonds and Self-Crosslinking. *Adv. Energy Mater.* **2025**, *15*, No. e03437.

(38) Chen, Q.; Ma, M.; Li, W.; Wang, Y.; Gao, Y.; Li, Y.; Liu, C. Improved Efficiency and Stability of Perovskite Solar Cells Through Long-Chain Phenylammonium Additives. *ACS Appl. Mater. Interfaces* **2025**, *17* (3), 5193–5201.

(39) Ma, J.; Lin, Z.; Guo, X.; Zhou, L.; Su, J.; Zhang, C.; Yang, Z.; Chang, J.; Frank Liu, S.; Hao, Y. Low-Temperature Solution-Processed ZnO Electron Transport Layer for Highly Efficient and Stable Planar Perovskite Solar Cells with Efficiency Over 20%. *Sol. RRL* **2019**, *3* (7), No. 1900096.